

Topic 1 — Independent Practice Part 2 Solutions

Brief checks. Use **We Need** → **We Know** → **We Solve** in your notebook.

Topic 1-1 Relate Integers and Opposites

Question 1

We Need: The additive inverse of a 3.5 m descent.

We Know: Additive inverse means same distance, opposite direction.

We Solve: Go 3.5 meters up (toward / above the starting level).

Answer: 3.5 meters upward

Question 2

We Need: An equation with $|p|$ and $|q|$.

We Know: $p + q = 0$ means $q = -p$, so the distances from zero match.

We Solve: $|p| = |q|$

Answer: $|p| = |q|$ (or equivalent)

Question 3

We Need: $|-11|$

We Know: Absolute value is distance from zero.

We Solve: $|-11| = 11$

Answer: 11

Question 4

We Need: Order -2 , 4 , 6 least to greatest.

We Know: On a number line, more left means smaller.

We Solve: -2 , 4 , 6

Answer: -2 , 4 , 6

Topic 1-2 Understand Rational Numbers

Question 1

We Need: Decimal for $\frac{5}{8}$.

We Know: Divide $5 \div 8$.

We Solve: $5 \div 8 = 0.625$

Answer: 0.625

Question 2

We Need: Which statement is *not* right.

We Know: A fraction (with nonzero denominator) becomes a decimal that terminates or repeats.

We Solve: B is not right.

Answer: B

Topic SC-1 Compare Rational Numbers

Question 1

We Need: Two comparisons of 9.3 and 9.8.

We Know: On a number line, more right means greater.

We Solve: $9.3 < 9.8$ and $9.8 > 9.3$

Answer: $9.3 < 9.8$ and $9.8 > 9.3$ (pounds OK)

Question 2

We Need: Symbols that fit $\frac{2}{5}$ and 0.4.

We Know: $\frac{2}{5} = 0.4$

We Solve: Equal numbers also satisfy $\leq$ and $\geq$.

Answer: $=$, $\leq$, $\geq$

Topic 1-3 Add Integers

Question 1

Part A: Start at 0. Move right 12, left 5, left 9.

Part B: $12 + (-5) + (-9) = -2$

Answer: Part A: number-line moves as above; Part B: -2

Question 2

We Need: The sum of the three signed moves.

We Know: Combine the positives and negatives.

We Solve: $18 + 4 + (-25) = -3$

Answer: -3

Question 3

We Need: Balance after 8 months.

We Know: Fee \$8.75 each month; $8 \times \$8.75 = \70 .

We Solve: $\$400 - \$70 = \$330$

Answer: \$330

Question 4

We Need: The sum.

We Know: Opposite signs: subtract the distances, keep the sign of the larger.

We Solve: $-8 + 14 = 6$

Answer: 6

Topic 1-4 Subtract Integers

Question 1

We Need: Distance from 12°F to 3°F .

We Know: Distance is the absolute value of the difference.

We Solve: $|12 - 3| = 9$. Valid: $|12 - 3|$, $|3 - 12|$, $12 - 3$.

Answer: Distance 9°F ; any correct distance expressions

Question 2

We Need: Distance between the two heights.

We Know: Distance is the absolute value of the difference.

We Solve: $|280 - 210| = 70$. Valid: $|280 - 210|$, $|210 - 280|$, $280 - 210$.

Answer: Distance 70 ft; any correct distance expressions

Question 3

We Need: Distance between the two depths.

We Know: Distance is the absolute value of the difference.

We Solve: $|-18 - (-45)| = 27$. Valid: $|-18 - (-45)|$, $|-45 - (-18)|$, $45 - 18$.

Answer: Distance 27 ft; any correct distance expressions

Topic 1-5 Add and Subtract Rational Numbers

Question 1

We Need: Balance after 5 months.

We Know: Start \$180 (below \$200), so the fee applies each month.

We Solve: $180 - 12 = 168$; $168 - 12 = 156$; $156 - 12 = 144$; $144 - 12 = 132$; $132 - 12 = 120$.

Answer: \$120

Topic 1-6 Multiply Integers

Question 1

Part A: Ascent is positive: 20 (or +20) feet per minute.

Part B: $20 \times 9 = 180$ feet above sea level.

Answer: Part A: 20 (or +20); Part B: 180 feet above sea level

Question 2

We Need: Height after 3.5 minutes.

We Know: An ascent rate is positive, so the product is positive.

We Solve: $180.25 \times 3.5 = 630.875$ m $= 0.630875$ km ≈ 1 km.

Answer: 1 kilometer (nearest km; ascent)

Question 3

We Need: $(-6)^2$

We Know: The parentheses mean the negative is part of the base.

We Solve: $(-6)^2 = (-6) \times (-6) = 36$

Answer: 36

Question 4

We Need: -6^2

We Know: Without parentheses, the exponent applies to 6 only.

We Solve: $-6^2 = -(6^2) = -36$

Answer: -36

Difference: parentheses square the negative; without them, square first, then make negative.

Topic 1-7 Multiply Rational Numbers

Question 1

We Need: The product.

We Know: Do the parentheses first, then multiply.

We Solve: $-0.2(15 + 5) = -0.2 \times 20 = -4$

Answer: -4

Topic 1-8 Divide Integers

Question 1

Part A: Unit rate = $140 \div 5 = 28$ feet per minute.

Part B: $28 \times 3 = 84$ feet above sea level.

Answer: Part A: 28 (or +28); Part B: 84 feet above sea level

Question 2

We Need: The quotient.

We Know: Opposite signs $\Rightarrow$ negative quotient.

We Solve: $(-81) \div 9 = -9$

Answer: -9

Question 3

We Need: The quotient.

We Know: Same signs $\Rightarrow$ positive quotient.

We Solve: $(-56) \div (-7) = 8$

Answer: 8

Question 4

We Need: n in $\frac{-24}{8} = \frac{n}{-3}$.

We Know: The fractions are equal. Use the butterfly method.

$$\begin{array}{cc} -24 & n \\ \hline & \\ \hline 8 & -3 \end{array}$$

Butterfly: multiply along each red line.

We Solve: Cross products: $(-24)(-3) = 8 \cdot n$, so $72 = 8n$ and $n = 9$.

Check: $(-24) \div 8 = -3$ and $9 \div (-3) = -3$.

Answer: $n = 9$

Topic 1-9 Divide Rational Numbers

Question 1

We Need: Each owner's share of the loss.

We Know: Split the loss equally among the 5 owners.

We Solve: $(-625.50) \div 5 = -125.10$

Answer: -125.10 (or $-\$125.10$ per owner)

Question 2

We Need: How many rest stops.

We Know: $16\frac{1}{2} = \frac{33}{2}$, $2\frac{3}{4} = \frac{11}{4}$.

We Solve: $\frac{33}{2} \div \frac{11}{4} = \frac{33}{2} \times \frac{4}{11} = 6$ intervals. Rest stops at the end of each interval (including the finish line) $\Rightarrow 6$.

Answer: 6 rest stops

Topic 1-10 Solve Problems with Rational Numbers

Question 1

We Need: The difference in elevation.

We Know: Subtracting a negative adds its opposite.

We Solve: $320 - (-85) = 320 + 85 = 405$

Answer: 405 feet

Topic SC-2 Use the Laws of Exponents**Question 1**

We Solve: $2^3 \times 2^5 = 2^{3+5} = 2^8 = 256$

Answer: 256 square inches

Question 2

Part A: $\frac{3^4}{3^2}$ or 3^{4-2} or 3^2

Part B: $81 \div 9 = 9$

Answer: Part A: $\frac{3^4}{3^2}$ (or equivalent); Part B: 9 cupcakes per friend

Question 3

We Solve: $(2^4)^2 = 2^{4 \cdot 2} = 2^8 = 256$

Answer: 256

Question 4

We Solve: $5^0 = 1$

Answer: 1

Question 5

We Solve: $\frac{5^5}{5^2} \cdot 5^0 \cdot 5^{-1} = 5^{5-2} \cdot 1 \cdot \frac{1}{5} = 5^3 \cdot \frac{1}{5} = 125 \div 5 = 25$

Answer: 25

Question 6

We Solve: $(3^2)^2 = 3^4 = 81$, $(2 \cdot 3)^2 = 6^2 = 36$, and $\left(\frac{18}{3}\right)^2 = 6^2 = 36$. Then $81 \cdot 36 \div 36 = 81$.

Answer: 81

Question 7

We Solve: $2^0 + 2^{-3} = 1 + \frac{1}{2^3} = 1 + \frac{1}{8} = 1\frac{1}{8}$

Answer: $\frac{9}{8}$ or $1\frac{1}{8}$

Question 8

We Solve: $\frac{10^5}{10^3} = 10^{5-3} = 10^2 = 100$

Answer: 100

Question 9

We Solve: $(4^2)^2 = 4^{2 \cdot 2} = 4^4 = 256$

Answer: 256

Question 10

We Solve: $(2 \cdot 3)^2 = 6^2 = 36$

Answer: 36

Question 11

We Solve: $\left(\frac{15}{3}\right)^2 = 5^2 = 25$

Answer: 25

Question 12

We Solve: $5^3 \cdot 5^2 = 5^{3+2} = 5^5 = 3125$

Answer: 3125